

US Candy Distribution Sales & Logistics Analytics

1. Project Overview and Objective:

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions. Analyze candy sales and distribution data to evaluate shipping efficiency, identify high-margin product lines, and recommend factory allocation improvements for optimizing supply chain performance.

2. Data Sources:

Source Description:	Timeline (2021-2024)
The dataset contains sales and geospatial shipment information for a US national candy distributor. It includes customer and factory location data, shipment routes, sales orders, product categories, sales goals, and margin-related information. The dataset is designed to analyze logistics efficiency, product profitability, and distribution optimization.	
Data Source: Maven Analytics	
Dataset Type: Public Domain Dataset	
Dataset Name: <i>US Candy Distributor</i>	
Data Collection Method: Simulated / business-case dataset provided by Maven Analytics for analytics and dashboard practice.	

Source Description:
Data Source: Maven Analytics – Public Dataset
Dataset provider: Maven Analytics
License: Public Domain
Dataset type: Sales + Geospatial + Logistics Data

3. Problem Statement:

Question 1
Which products have the lowest sales? Fun Dip (\$12.00), Nerds (\$15.00), Laffy Taffy (\$53.73), SweeTARTS (\$61.50), Hair Toffee (\$76.50)
Visual
Clustered Bar Chart
Axis
Product Name
Values
Total Sales

Question 2
Which products contribute the most to total sales?

- Wonka Bar - Triple Dazzle Caramel (\$28,485.00)
- Wonka Bar - Scrumdiddlyumptious (\$27,874.80)
- Wonka Bar - Milk Chocolate (\$26,867.75)
- Wonka Bar - Fudge Mallows (\$24,890.40)
- Wonka Bar - Nutty Crunch Surprise (\$23,574.95)

Visual

Clustered Bar Chart

Axis

Product Name

Values

Total Sales

Sort **Descending**.

Top products appear first.

You can also apply

Top N Filter

Top 10

By Total Sales

Question 3

Which product lines have the highest profit margins?

Sugar – 68.10%
 Chocolate – 67.45%
 Other – 44.84%

Visual

Bar Chart

Axis

Product Line

Values

Profit Margin %

Sort Descending.

Highest percentage = Highest profit margin.

Question 4

Which months recorded the highest and lowest sales performance?

Highest: December (\$21,215.73)

Lowest: February (\$3,933.52)

Visual

Line Chart

Axis

Order Date → Month

Values

Total Sales

Sort by Month Number.

Highest point

Highest sales month

Lowest point

Lowest sales month

Question 5

Which product categories generate the highest total sales revenue?

- Chocolate – \$131,692.90
- Other – \$9,663.25
- Sugar – \$348.73

Visual

Column Chart

Axis

Category

Values

Total Sales

Sort Descending.

Highest column

Highest revenue category.

Question 6

Which locations have the lowest sales performance?

Melbourne (\$2.50), Littleton (\$3.60), Jupiter (\$3.60), Iowa City (\$3.60), Port Orange (\$3.60)

Visual

Map

Axis

City / Location

Values

Total Sales

Sort Ascending.

Lowest locations appear first.

Question 7

Which states or regions have the highest sales and profit?

California leads with Sales: \$27,917.40 and Gross Profit: \$18,479.42,
followed by New York and Texas.

Visual

Filled Map

or

Clustered Column Chart

Axis

State

Values

- Total Sales
- Total Profit

Sort by Sales.

Highest states appear first.

Question 8

Which product categories exceeded their sales targets?

Chocolate (Sales: \$131,692.90 vs Target: \$27,000)

Other (Sales: \$9,663.25 vs Target: \$3,000)

Sugar did not meet its target (Sales: \$348.73 vs Target: \$15,000).

Create Measure

Target Difference = [Total Sales]-[Sales Target]

Visual

Table

Columns

- Category
- Total Sales
- Sales Target
- Target Difference

Filter

Target Difference > 0

These categories exceeded the target.

Question 9

Which customers generate the highest sales revenue?

Customer IDs:

- 131807 (\$454.40),
- 115238 (\$380.50),
- 122336 (\$352.20),
- 164756 (\$304.14),
- 124163 (\$247.35)

Visual

Bar Chart

Axis

Customer

Values

Total Sales

Sort Descending.

Top customer = Highest revenue.

Question 10

Which factories manufacture the highest-selling products?

- Lot's O' Nuts – \$76,340.15
- Wicked Choccy's – \$55,352.75
- Secret Factory – \$8,587.50

Visual

Matrix

Rows

Factory

Columns

Product Name

Values

Total Sales

Question 11

Which factories contribute the highest sales and profit?

Lot's O' Nuts contributes the highest sales, followed by Wicked Choccy's. These factories also generate the highest profits because they manufacture the best-selling product lines.

Visual

Clustered Bar Chart

Axis

Factory

Values

- Total Sales
- Total Profit

Sort by Sales.

Highest factories appear first.

Question 12

Which product divisions contribute the most revenue?

Chocolate (\$131,692.90)
Other (\$9,663.25)
Sugar (\$348.73)

Visual

Column Chart

Axis

Division

Values

Total Sales

Sort Descending.

Top division = Highest revenue.

Question 13

Which states receive the highest number of orders?

California – 2,001 orders
New York – 1,127 orders
Texas – 985 orders
Pennsylvania – 586 orders
5. Washington – 506 orders

Create Measure

Order Count = `DISTINCTCOUNT('Candy Sales'[Order ID])`

Visual

Column Chart

Axis

State

Values

Order Count

Sort Descending.

Highest state = Most orders.

4. Attribute Details:

5. Tools & Technologies

Tools:

- **Excel:** Data cleaning, data transformation, Pivot Tables, data analysis, and reporting.
- **Power BI:** Data modeling, DAX calculations, interactive dashboard development, and data visualization.

Technologies:

Field	Data Type	Description
Row ID	Integer / String	Unique row identifier
Order ID	Integer / String	Unique order identifier
Order Date	Categorical	Date of order
Ship Date	Categorical	Date of shipment
Ship Mode	Integer / String	Shipping method of order
Customer ID	Integer / String	Unique customer identifier
Country/Region	String (Text)	Country or region of customer
City	String (Text)	City of customer
State/Province	String (Text)	State/province of customer
Postal Code	Numeric (Integer)	Postal code / zip code of customer
Division	Categorical	Product division
Region	String (Text)	Region of customer
Product ID	String (Text)	Unique product identifier
Product Name	String (Text)	Product long name
Sales	Numeric (Integer)	Total sales value of order
Units	Numeric (Integer)	Total units of order
Gross Profit	Numeric (Integer)	Gross profit of order (Sales - Cost)

Cost	Numeric (Integer)	Cost to manufacture
Factory	String (Text)	Factory Name
Latitude	Numeric (Integer)	Latitude Coordinates
Longitude	Numeric (Integer)	Longitude Coordinates
Division	String (Text)	Product Division
Product Name	String (Text)	Product Descriptive Name
Factory	Categorical	Factory Name
Product ID	Integer / String	Product Unique Identifier
Unit Price	Numeric (Integer)	Selling Price of Product
Unit Cost	Numeric (Integer)	Cost to Produce Product
Division	Categorical	Product Division
Target	String (Text)	Sales Target 2024
zip	Numeric (Integer)	The 5-digit zip code assigned by the U.S. Postal Service.
lat	Numeric (Integer)	The latitude of the zip code (learn more).
lng	Numeric (Integer)	The longitude of the zip code (learn more).
city	String (Text)	The official USPS city name.
state_id	Numeric (Integer)	The official USPS state abbreviation.
state_name	String (Text)	The state's name.
zcta	Integer / String	TRUE if the zip code is a Zip Code Tabulation area (learn more).
parent_zcta	String (Text)	The ZCTA that contains this zip code. Only exists if zcta is FALSE. Useful for making inferences about a zip codes that is a point from the ZCTA that contains it.
population	String (Text)	An estimate of the zip code's population. Only exists if zcta is TRUE.
density	String (Text)	The estimated population per square kilometer. Only exists if zcta is TRUE.

county_fips	Integer / String	The zip's primary county in the FIPS format.
county_name	Integer / String	The name of the county_fips.
county_weights	Numeric (Integer)	A JSON dictionary listing all county_fips and their weights (by area) associated with the zip code.
imprecise	String (Text)	TRUE if the lat/lng has been geolocated using the city (rare).
military	Numeric (Integer)	TRUE if the zip code is used by the US Military (lat/lng not available).

- **Data Analysis Techniques:** KPI tracking, trend analysis, performance analysis, and business insights generation..

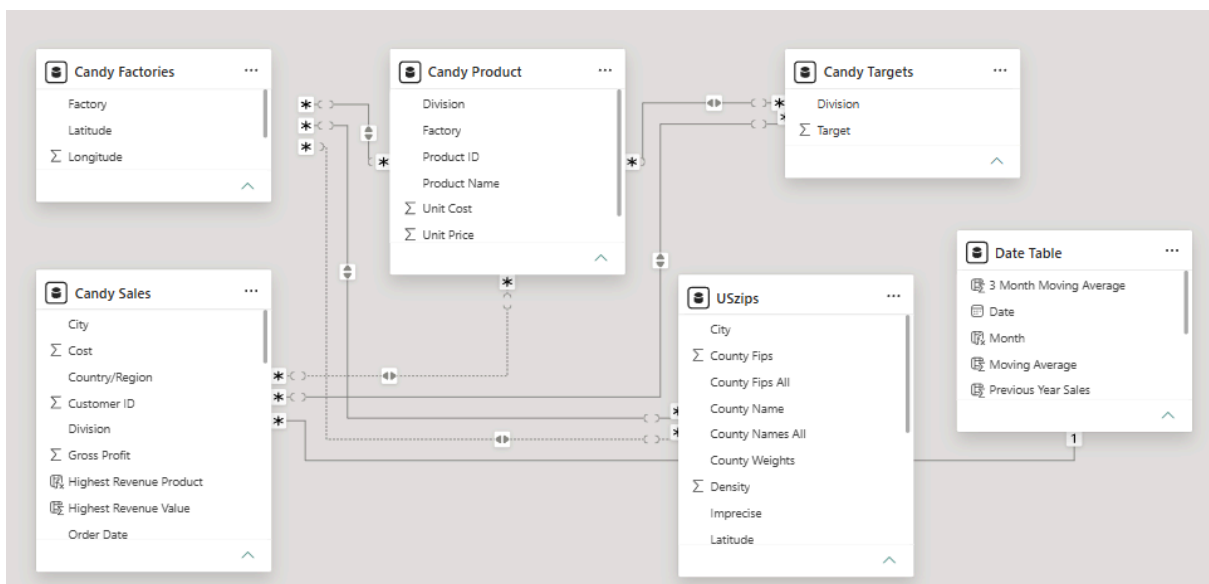
6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



SCREENSHOT OF DATA MODELLING

- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.

1. Average Orders per Factory = `AVERAGEX`

```

(
    VALUES('Candy Factories'[Factory]),
    [Total Orders]
)
2.    Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
3.    Factory Rank =
RANKX(
    ALL('Candy Product'[Factory]),
    CALCULATE([Total Sales]),
    ,
    DESC,
    DENSE
)
4.    Average Revenue = AVERAGE('Candy Sales'[Sales])
5.    Goal Achievement % =
DIVIDE(
    [Total Actual Sales],
    [Total Sales Goal],
    0
)
6.    Highest Revenue Product =
VAR TopProduct =
    TOPN(
        1,
        SUMMARIZE(
            'Candy Sales',
            'Candy Sales'[Product Name],
            "Sales", [Total Sales]
        ),
        [Sales],
        DESC
    )
RETURN
    MAXX(TopProduct, 'Candy Sales'[Product Name])
7.    Highest Revenue Value =
MAXX(
    ADDCOLUMNS(
        ALL('Candy Sales'[Customer ID]),
        "Revenue", CALCULATE([Total Sales])
    ),
    [Revenue]
)
8.    Order Count = DISTINCTCOUNT('Candy Sales'[Order ID])
9.    Performance Status =
IF(
    [Total Actual Sales] >= [Total Sales Goal],
    "Goal Achieved",
    "Below Goal"
)

```

10. Product Rank =

```
RANKX(  
    ALL('Candy Sales'[Product Name]),  
    [Total Sales],  
    ,  
    DESC,  
    DENSE  
)
```

11. Profit = SUM('Candy Sales'[Sales]) - SUM('Candy Sales'[Cost])

12. Region Rank =

```
RANKX(  
    ALL('Candy Sales'[Region]),  
    [Total Sales],  
    ,  
    DESC,  
    DENSE  
)
```

13. Sales Target = SUM('Candy Targets'[Target])

14. Sales Variance = [Total Actual Sales] - [Total Sales Goal]

15. Target Difference = [Total Sales] - [Sales Target]

16. Target Status =

```
IF(  
    [Target Difference] > 0,  
    "Exceeded Target",  
    "Below Target"  
)
```

17. Total Actual Sales = SUM('Candy Sales'[Sales])

18. Total Customers = DISTINCTCOUNT('Candy Sales'[Customer ID])

19. Total Orders = COUNTROWS('Candy Sales')

20. Total Products = DISTINCTCOUNT('Candy Sales'[Product Name])

21. Total Profit =

```
SUMX(  
    'Candy Sales',  
    'Candy Sales'[Sales] - 'Candy Sales'[Cost]  
)
```

Name: Date Table

Structure: Manage relationships, Calculations, Calendars

Report view: Date Table =

Date	Year	Month	Quarter	Year Month	Previous Year Sales	YoY Growth %	Running Sales	Moving Average	3 Month Moving Average
03-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
04-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
05-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
06-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
07-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
08-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
09-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
10-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
11-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
12-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
13-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
14-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
15-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
16-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
17-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
18-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
19-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
20-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
21-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
22-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
23-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
24-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49
25-Jan-21 12:00:00 AM	2021	January	Q1	Jan 2021			₹ 141,783.63	₹ 206.4984	₹ 206.49

Table: Date Table (1,458 rows)

Data: Candy Factories, Candy Product, Candy Sales, Candy Targets, Division, Target, Date Table, 3 Month Moving Average, Date, Month, Moving Average, Previous Year Sales, Quarter, Running Sales, Year, Year Month, YoY Growth %, USZips

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statements:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**
- Make your Report interactive Layout using **Drill-down, Filters and slicers**
- **Bookmarks, Clear Titles for each visuals & Labels is must.**
- **Create a consolidated Report /Dashboard.**

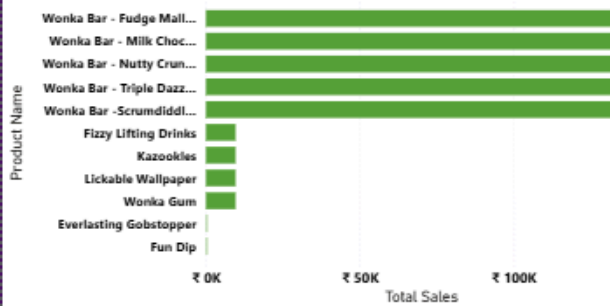
SCREENSHOT OF DASHBOARD /REPORT

1. Product Analysis:

US Candy Distribution Sales & Logistics Analytics - Product Analysis

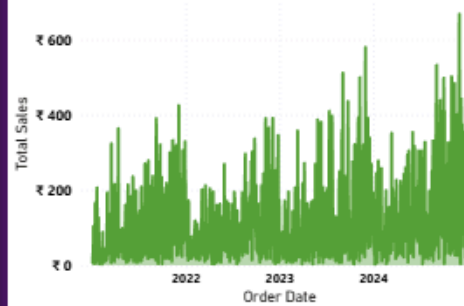
Total Sales by Product Name and Highest Revenue Product

Highest Revenue Product ● Wonka Bar - Triple Dazzle Caramel



Total Sales by Order Date and 3 Month Moving Average

3 Month Movin... ● ₹ 206,4984

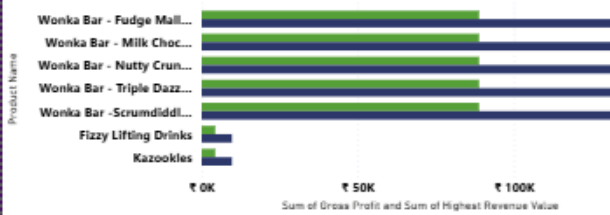


Month

- ☐ Select all
- ☐ April
- ☐ August
- ☐ December
- ☐ February
- ☐ January
- ☐ July
- ☐ June
- ☐ March
- ☐ May
- ☐ November

Sum of Gross Profit and Sum of Highest Revenue Value by Product Name

● Sum of Gross Profit ● Sum of Highest Revenue Value

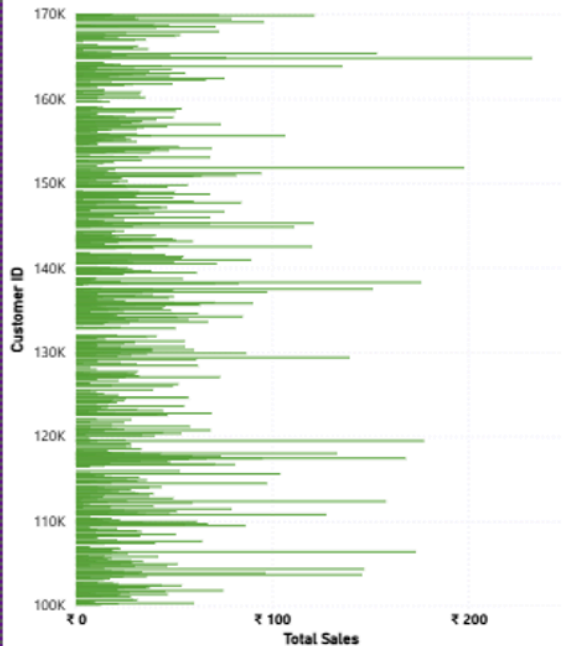


Factory	Total Actual Sales	Total Sales Goal	Goal Achievement %	Performance Status
Lot's O' Nuts	₹ 131,692.9	₹ 131,692.9	1.00	Goal Achieved
Secret Factory	₹ 10,090.73	₹ 10,090.73	1.00	Goal Achieved
Sugar Shack	₹ 10,090.73	₹ 10,090.73	1.00	Goal Achieved
The Other Factory	₹ 10,090.73	₹ 10,090.73	1.00	Goal Achieved
Wicked Choccy's	₹ 131,692.9	₹ 131,692.9	1.00	Goal Achieved
Total	₹ 141,783.63	₹ 141,783.63	1.00	Goal Achieved

2. Customer Analysis:

US Candy Distribution Sales & Logistics Analytics - Customer Analysis

Total Sales by Customer ID



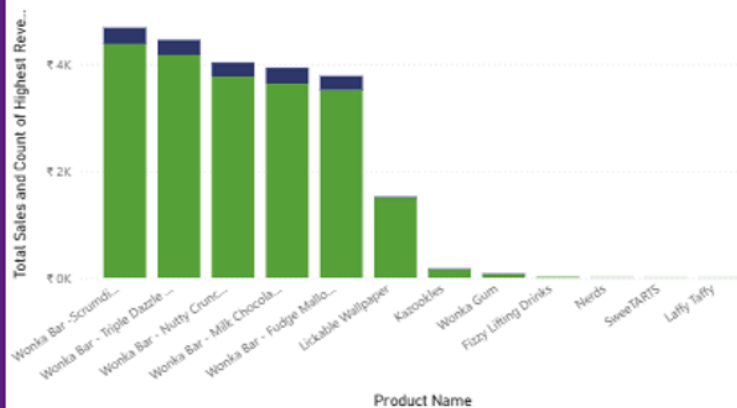
Month

- ☒ Select all
- ☐ April
- ☐ August
- ☒ December
- ☐ February

Division	Total Sales	Sales Target	Target Difference
Chocolate	₹ 19,474.5	\$27,000	(\$7,525.5)
Other	₹ 1,726.75	\$3,000	(\$1,273.25)
Sugar	₹ 33.23	\$15,000	(\$14,966.77)
Total	₹ 21,234.48	\$45,000	(\$23,765.52)

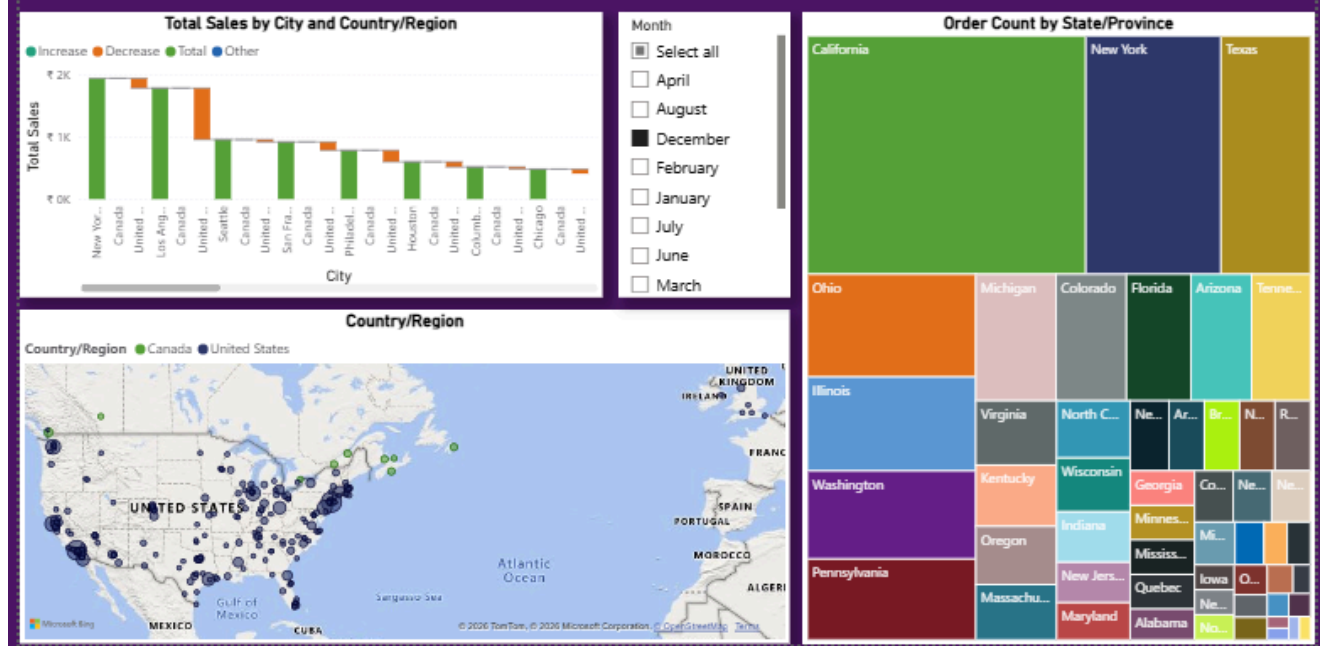
Total Sales and Count of Highest Revenue Product by Product Name

● Total Sales ● Count of Highest Revenue Product



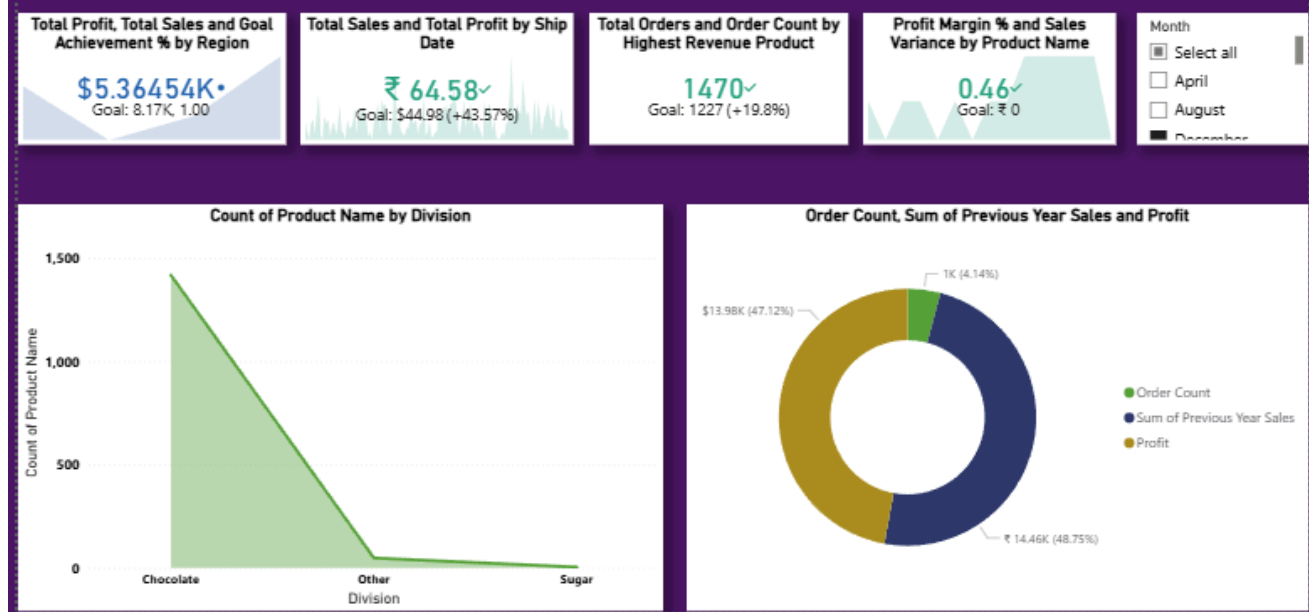
3. Geography Analysis:

US Candy Distribution Sales & Logistics Analytics - Geography Analysis



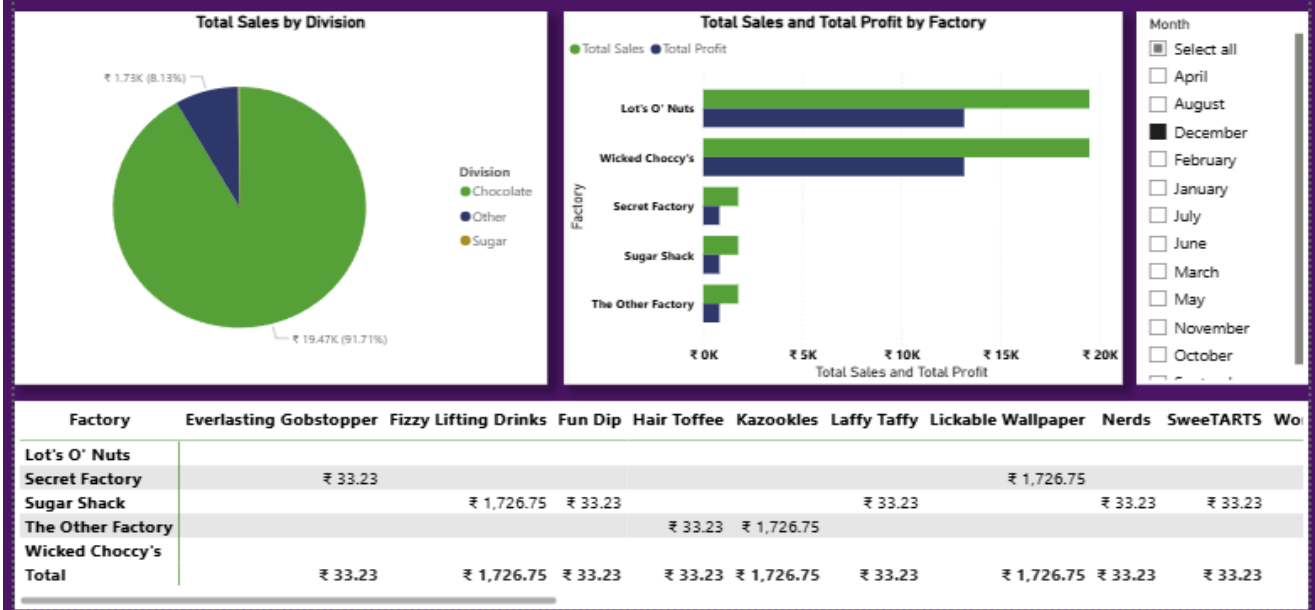
4. KPIs:

US Candy Distribution Sales & Logistics Analytics - KPIs



5. Factory Analysis:

US Candy Distribution Sales & Logistics Analytics - Factory Analysis



9. Insights & Conclusions

- **Key Findings:**
- **Chocolate** is the dominant product division, contributing the vast majority of revenue.
- **Wonka Bar - Triple Dazzle Caramel** is the top-selling product.
- **Lot's O' Nuts** is the highest-performing factory.
- **California** is the strongest market in terms of sales, profit, and order volume.
- **December** is the peak sales month, while **February** records the lowest sales.
- **Chocolate** and **Other** divisions exceeded their sales targets, whereas **Sugar** significantly underperformed.
- Chocolate products, California, and Lot's O' Nuts generate the highest sales.
- Revenue is driven by a small number of high-performing products, factories, and states.
- Historical trends indicate continued strong demand for Chocolate products and key markets.
- Increase production of top-selling products, optimize factory allocation, strengthen high-performing markets, and improve low-performing categories.
- **Provide the analysis insights:**

Chocolate is the leading product division, generating approximately ₹131.7K in sales, with the Wonka product line contributing the highest revenue. California is the top-performing state, while Lot's O' Nuts and Wicked Choccy's are the highest-selling factories. Sales remain relatively stable over time, indicating consistent demand, particularly for Chocolate products. To improve business performance, the company should increase production of high-demand Wonka products, strengthen inventory at top-performing factories, promote low-selling Sugar products, focus on expanding in high-performing regions, and review underperforming products for improvement or discontinuation.

10. Conclusions:

The **Candy Distribution Sales & Logistics Analytics Dashboard** provides a comprehensive view of sales performance, product profitability, customer demand, geographic trends, and factory contributions. By leveraging Power BI's interactive visualizations and DAX measures, the dashboard transforms raw sales data into meaningful business insights.

The analysis reveals that **Chocolate** is the highest-performing product division, contributing the majority of total revenue, while **Wonka Bar** products are the top-selling products across the portfolio. **California, New York, and Texas** generate the highest sales and order volumes, highlighting key markets for the business. Factory performance analysis shows that **Lot's O' Nuts** and **Wicked Choccy's** contribute the highest sales and profit, indicating their importance in overall business performance. Additionally, comparing actual sales with targets helps identify product categories that exceeded expectations and those requiring improvement.

Overall, this dashboard enables stakeholders to monitor key performance indicators (KPIs), identify high- and low-performing products and regions, evaluate factory performance, and support data-driven decision-making. These insights can help optimize sales strategies, improve resource allocation, and drive future business growth.